County geography adjustments inside the CH index

Controls: CH specification—County-adjusted CH; Geography; County calibration instrument; Include log county area in the calibration.

1. Estimate geography effects using counties. This option estimates geography effects where there are many county observations, then incorporates those effects into the MSA or state index. For county c, let q_c be all-industry GDP, n_c total employment, a_c land area and $d_c = n_c/a_c$ density. The linked county regression is

$$\log(q_c/n_c) = \alpha_{s(c)} + \rho_C \log d_c + X_c'\gamma + \lambda \log a_c + u_c.$$

Here $\alpha_{s(c)}$ is a modern state fixed effect, X_c contains the selected geography and transport variables, γ their coefficients, and u_c the error. Log county area is optional and off by default. Its coefficient is λ. The county density coefficient is ρ_C.

2. The calibration instrument and sample. With IV, historical county population in levels instruments log density. The population construction can be selected separately from the regional instrument. The county regression uses the same historical sample year, instrument year and territory scope as the regional model. With OLS, it estimates the same county equation without an instrument.

Calibration uses eligible counties in all states represented by the retained regions, including nonmetropolitan counties. It therefore need not reproduce a standalone County IV result based on a different sample. County slopes agree when inputs, samples, controls and fixed effects agree. The geography variables are treated as exogenous controls.

3. Transfer geography effects into the index. For region g, total employment is $N_g = \sum_{c \in g} n_c$. Construct

$$D_g^{\text{geo}}(\theta) = \frac{1}{N_g} \sum_{c \in g} \exp(X_c'\widehat{\gamma})\, n_c^{\theta} a_c^{1-\theta}.$$

Only the estimated geography and transport effects enter the multiplier. County fixed effects, the optional area coefficient and ρ_C are not transferred. The regional regression relates log GDP per employment to $\log D_g^{\text{geo}}(\theta)$ and estimates its own elasticity $\rho = \theta - 1$. In MSA analyses, schooling and college can still adjust the regional outcome separately.

4. Missing inputs and uncertainty. Every constituent county remains in the index, even if its historical instrument is unavailable for calibration. A missing required input in a positive-employment county excludes the entire region. Joint standard errors account for estimated geography coefficients and observations shared by the county and regional regressions. This requires the county calibration to be identified. Using county estimates reduces the number of effects estimated from states; it does not remove the assumptions needed to transfer geography coefficients across counties and regions.